

● **UrbanPulse AI**

UrbanPulse AI

An Agentic Urban Intelligence Platform for Smarter Cities

A multi-agent decision support system designed to automate grievance triage, deduplicate spatial events, calculate ward-level health analytics, and forecast infrastructure failure points.

SUBMISSION CONTEXT

Lenovo LEAP AICTE Hackathon 2.0

FOCUSED AREA THEME

 Smart Cities & Public Services

TEAM MEMBERS

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SUBMISSION DATE

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1. Executive Summary

Urban centers globally face an influx of population, straining public infrastructure and civic maintenance departments. Modern municipal corporations receive thousands of daily reports regarding road degradation, illegal refuse dumping, pipeline fractures, and malfunctioning grid utilities. However, existing municipal solutions are passive—acting as database storage queues that require manual sorting, suffer from report duplication, and offer zero analytical forecasting.

UrbanPulse AI is an autonomous, multi-agent Agentic Urban Intelligence Platform designed to transform this domain. By orchestrating a collaborative team of specialized AI agents built on top of stateful workflow graphs, the platform ingests multi-modal inputs (images, vernacular voice memos, and text) directly from citizens. It filters duplicate reports, computes context-aware severity indexes, routes tickets to correct departments, performs spatial trend predictive analysis, and monitors resolution SLAs. UrbanPulse AI transitions city management from reactive damage triage to proactive, predictive maintenance, maximizing resource efficiency and restoring citizen trust.

2. Project Objective

The primary objectives of the UrbanPulse AI platform are:

- **Automated Triage & Routing:** Target the elimination of manual sorting backlogs by leveraging LLM semantic routing, directing complaints to the correct department within seconds of ingestion.
- **Ticket Deduplication:** Implement smart spatial and visual grouping to consolidate overlapping citizen reports into a single master incident hotspot to preserve municipal database bandwidth.
- **Dynamic Severity Calculation:** Shift from first-in, first-out ticketing to community-impact prioritization by dynamically scoring incidents based on proximity to infrastructure (schools, hospitals, main arteries).
- **Proactive Failure Mitigation:** Leverage spatial analytics and telemetry anomaly correlation to flag early-stage infrastructure weaknesses (e.g., minor water leaks indicating pipe fatigue) before catastrophic failure occurs.
- **Vernacular Accessibility:** Integrate regional speech-to-text pipelines to allow elderly, rural, or illiterate populations to report issues natively.

3. Problem Statement & Need for the Solution

Indian cities are rapidly digitizing, but municipal operational backends remain archaic. When a citizen identifies a civic issue, the friction to report and resolve it is high:

1. **Departmental Silos:** Citizens do not know if a leaking pipe falls under the Water Supply Department, the Road Maintenance Wing, or the Sanitation Board, leading to incorrectly routed tickets.
2. **Duplicate Inundation:** A public hazard on a major lane generates dozens of separate tickets, causing administrative bloat and department fatigue.
3. **Reactive Fixes:** Cities spend major budgets on emergency, high-cost repairs because they lack the intelligence to correlate minor, repeated reports of leakage or small cracks.

4. **Lack of Transparency:** Citizens rarely receive status updates, fostering public distrust in municipal agencies.

There is an urgent need for an intelligent system that acts as an autonomous intermediary, converting raw citizen reports into parsed, prioritized, deduplicated, and trackable city maintenance workorders.

4. Existing Challenges (Why Current Portals Fail)

Existing citizen grievance portals suffer from structural failures:

- **Passive Databases:** They are static forms that dump text into PostgreSQL tables without verifying the images or checking the validity of the claims.
- **Manual Routing Vulnerability:** They rely on human operators to review each ticket, leading to backlogs and misclassifications.
- **No Spatial Awareness:** They treat each report as an isolated island, failing to group close-proximity reports.
- **Static SLA Trackers:** Service Level Agreements are static countdown timers that do not escalate issues, alert upper-management, or adjust urgency dynamically.

5. Proposed Solution: UrbanPulse AI

UrbanPulse AI is an **Agentic Urban Intelligence Platform** that serves as the cognitive backend of city operations. The platform consists of a citizen mobile web client for report capturing (featuring direct camera integration, GPS pinning, and voice recording) and a municipal dashboard for administration.

In between these sits the **UrbanPulse Agent Engine**, a stateful multi-agent system built on top of LangGraph. This engine models the entire lifecycle of an incident as a state graph, where specialized agents perform discrete operations (classification, deduplication, prioritization, department assignment, trend aggregation, and escalation) and update the incident state statefully.

 **Platform Security & Data Privacy**

UrbanPulse AI implements secure role-based access control (RBAC) via Supabase Auth for municipal staff and citizens. The Vision Agent automatically runs redaction algorithms to blur faces and private license plates in uploaded photos before they are stored in encrypted Supabase Storage buckets. Furthermore, all agent state transitions are logged in an append-only database audit trail to prevent ticket tampering.

FEATURE	LEGACY PORTAL (WEB2)	GENAI CHATBOT (1.0)	URBANPULSE AI (AGENTIC)
Ingestion Mode	Rigid text forms only	Chat prompt translation	Multi-modal (Vision + Voice + GIS)
Triage & Routing	100% Manual Human sorting	Keyword-based routing	Autonomous Department Agents
Duplicate Reports	Stored as separate records	Must be flag-checked manually	Smart GIS Hotspot Aggregation

FEATURE	LEGACY PORTAL (WEB2)	GENAI CHATBOT (1.0)	URBANPULSE AI (AGENTIC)
Urgency Scoring	Static (first-come, first-serve)	Basic keyword severity	Proximity & Context-aware priorities
Action Cycle	Passive ticket collection	Notification delivery	SLA tracking & Auto-Escalation

6. Multi-Agent Architecture

The core intelligence of UrbanPulse AI is driven by a stateful multi-agent network orchestrated via LangGraph.

Why LangGraph? LangGraph was selected because it enables stateful orchestration, conditional routing, automatic retries, and coordination between specialized AI agents. Unlike standard linear pipelines, LangGraph allows *cycles & loops* (e.g. if the Vision Agent finds an uploaded image incomplete, it routes the state back to the Citizen Experience Agent to ask for a clearer photo) and maintains a unified database state schema.



7. Detailed AI Agent Responsibilities

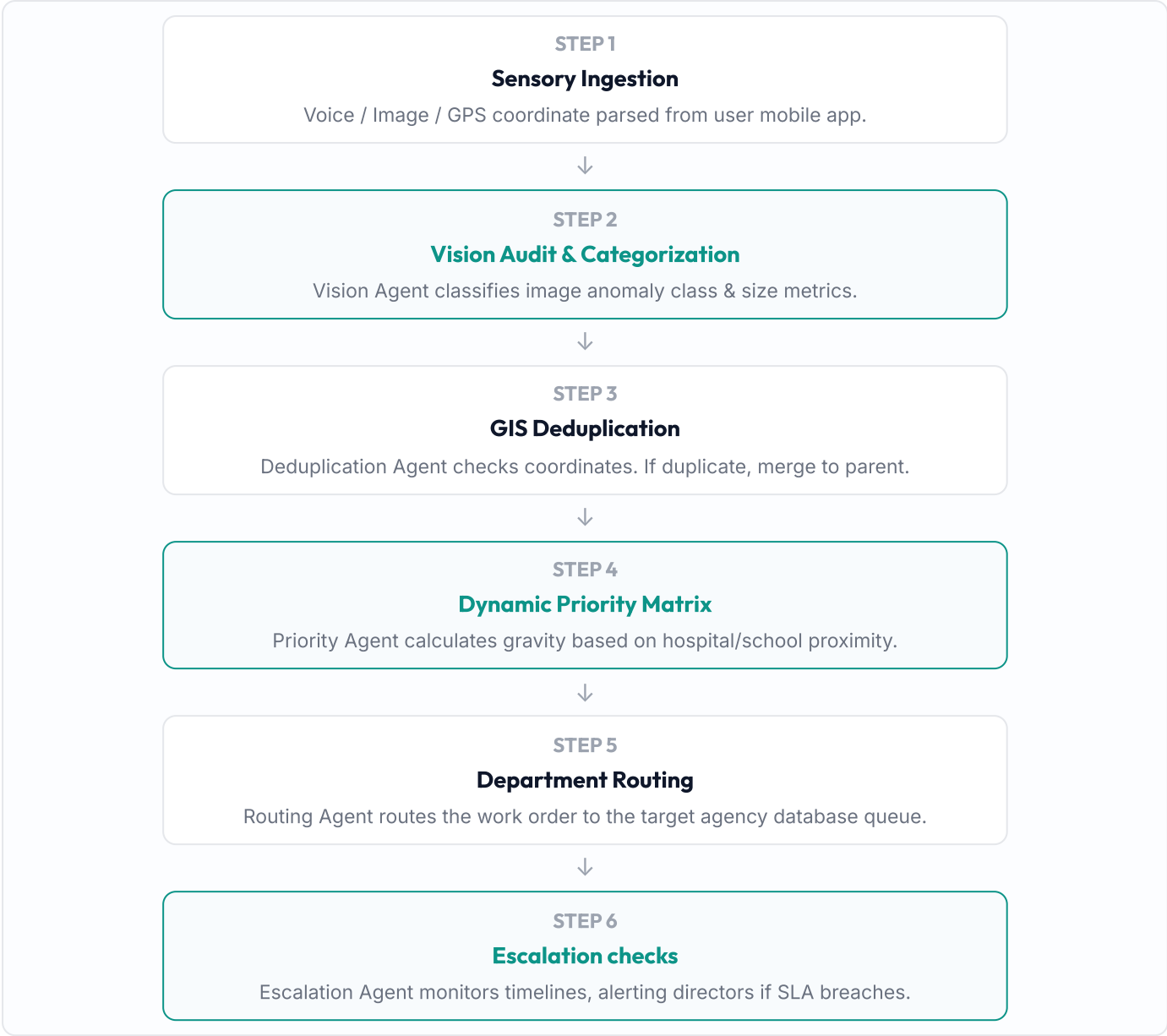
The operational responsibilities, inputs, and outputs of the specialized AI agents are detailed in the table below:

AGENT	RESPONSIBILITY	PRIMARY INPUTS	PRIMARY OUTPUTS
Vision Agent	Examines photos, detects anomaly categories, and estimates physical dimensions.	Base64 Image, Timestamp	Anomaly Category, Confidence Score, Dimensions
Deduplication Agent	Matches GPS coordinates and image embeddings against active database tickets.	Coordinates, Category, Image Embeddings	IsDuplicate (Boolean), Parent Incident ID

AGENT	RESPONSIBILITY	PRIMARY INPUTS	PRIMARY OUTPUTS
Priority Agent	Computes gravity score based on spatial layers (schools, hospitals, road types).	Base Severity, Proximity Metadata	Priority Score (0-100), Priority Level
Routing Agent	Classifies and directs the ticket to the correct department queue.	Category, Text Description	Target Department, Queue ID
Analytics Agent	Updates the Urban Health Score and generates AI City Pulse recommendations.	Active incident counts, Ward IDs, SLAs	Ward Health Score, Anomaly Trends, Actions
Citizen Experience (CX)	Handles translation, composes SMS/WhatsApp dispatches, and requests closure photos.	Ticket Status, Selected Language	Outbound Message Content
Escalation Agent	Monitors open ticket ages against SLA timers and alerts supervisors.	Incident ID, Priority Level, Elapsed Hours	IsEscalated (Boolean), Escalation Target

8. System Workflow & Decision Routing

The processing of citizen feedback is managed as a state machine where ticket states are evaluated at each agent node.



9. Key Platform Features & Differentiators

UrbanPulse AI stands out from simple ticket managers through four key features:

1. Automated Deduplication Engine

Clustered reports prevent database bloat. Duplicate reports boost the parent incident's dynamic priority, letting operators know which issues are affecting the most citizens.

2. Ward Urban Health Score (UHS)

Computes ward scores dynamically (0-100). If a ward's score drops below critical thresholds, resource allocation planners are notified to schedule preventative inspections.

3. AI City Pulse (Signature Recommendation Engine)

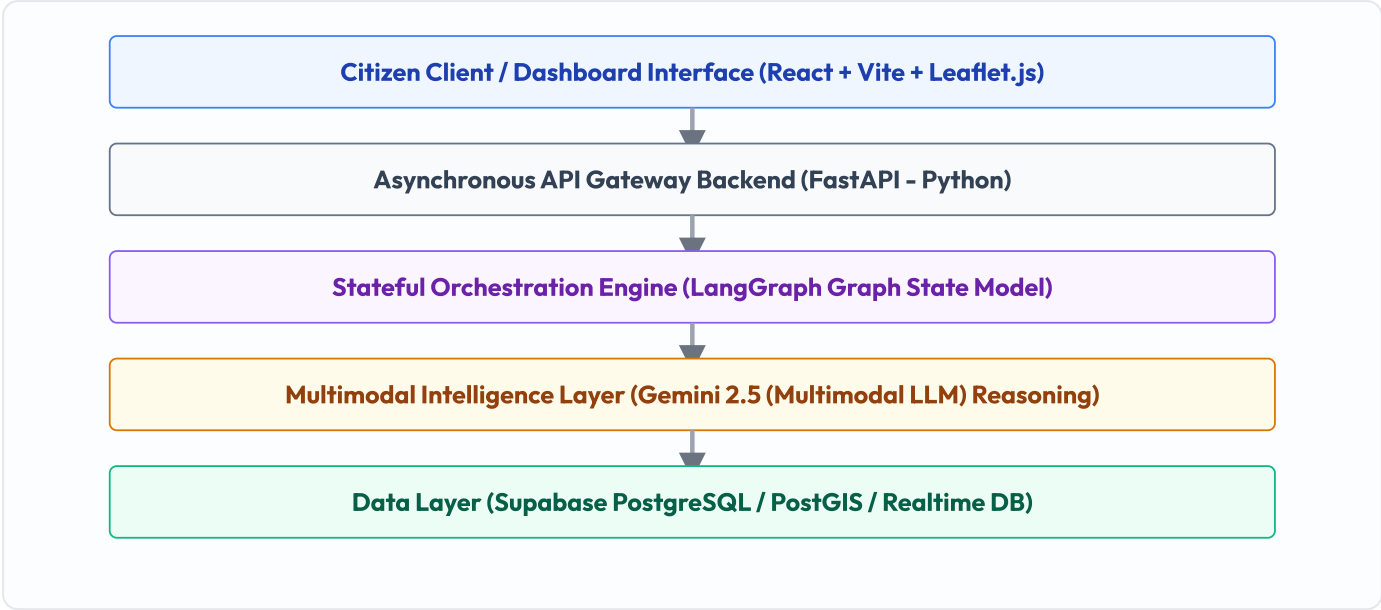
Translates raw spatial database metrics into plain-text trending briefs and actionable recommendations. E.g. *"Ward 12 Health: UHS 42 (Critical). Trending Issues: Water Leakage ↑ (4 reports). Recommended Action: Dispatch water repairs team to inspect pipes at coordinates #904."*

4. Predictive Trend Analysis

Aggregates historical report coordinates. If minor reports spike in a specific coordinate grid, the system alerts planners to investigate pipe or road wear before failures occur.

10. Technology Stack & Layered Architecture

The platform architecture is built on the following technologies:



11. Expected Impact & Value Proposition

Metric Category	Traditional Methods	With UrbanPulse AI Target	Operational Gain
Triage & Dispatch Delay	12 - 36 Hours (Manual)	Automated, instant dispatch	Reduced Triage Latency
Grievance Backlog Volume	High duplicate ticket clutter	Clean parent-child merging	Deduplication Target

Metric Category	Traditional Methods	With UrbanPulse AI Target	Operational Gain
Emergency Infrastructure Costs	High reactive repair expenses	Preventative scheduled work	Budget Planning Efficiency
Citizen SLA Compliance Rate	Variable SLA compliance	Escalations ensure faster loop	Improved Resolution Rate

12. Development Roadmap

The implementation strategy is divided into the current MVP achievements and future roadmap expansions:

Current MVP Implementation

- **Citizen Web App Prototype:** Mobile-responsive client supporting photo attachments and Leaflet.js map coordinate pinning.
- **FastAPI Backend Skeleton:** Simple ingestion API interfaces and mock datastore routers.
- **LangGraph Workflows:** Graph state routing code coordinating loops between Mock Vision and Department Routing nodes.
- **Command Dashboard:** Basic municipal operator cockpit displaying mock active tickets and Active Agent logs.

Planned Features

- **Predictive Analytics Engine:** Analytics Agent module tracking coordinate clusters and historical trends to predict grid failures.
- **IoT & Telemetry Feed Integration:** Syncing grid-telemetry from water pressure networks and smart streetlights.
- **Drone Survey Paths:** Integrating autonomous drone dispatch schemas over high-risk coordinates.
- **Satellite Vision Pipeline:** Visual analysis of satellite feeds to monitor large-scale municipal wear.

13. Risks & Limitations

- **Visual Edge Cases:** Insufficient lighting, extreme camera angles, or shadows may cause the Vision Agent to misinterpret anomalies.
- **GPS Drift Inaccuracies:** High-density urban areas with tall skyscrapers can lead to GPS inaccuracies, causing minor coordinates errors in deduplication boundaries.
- **Legacy System API Incompatibility:** Success depends on municipal backends having API hook access points.
- **Connectivity Reliance:** The system is dependent on active internet connections for real-time mobile reporting and database syncs.
- **False Deduplication Grouping:** Close-proximity incidents of identical category (e.g. two separate potholes 5 meters apart) may be incorrectly merged.

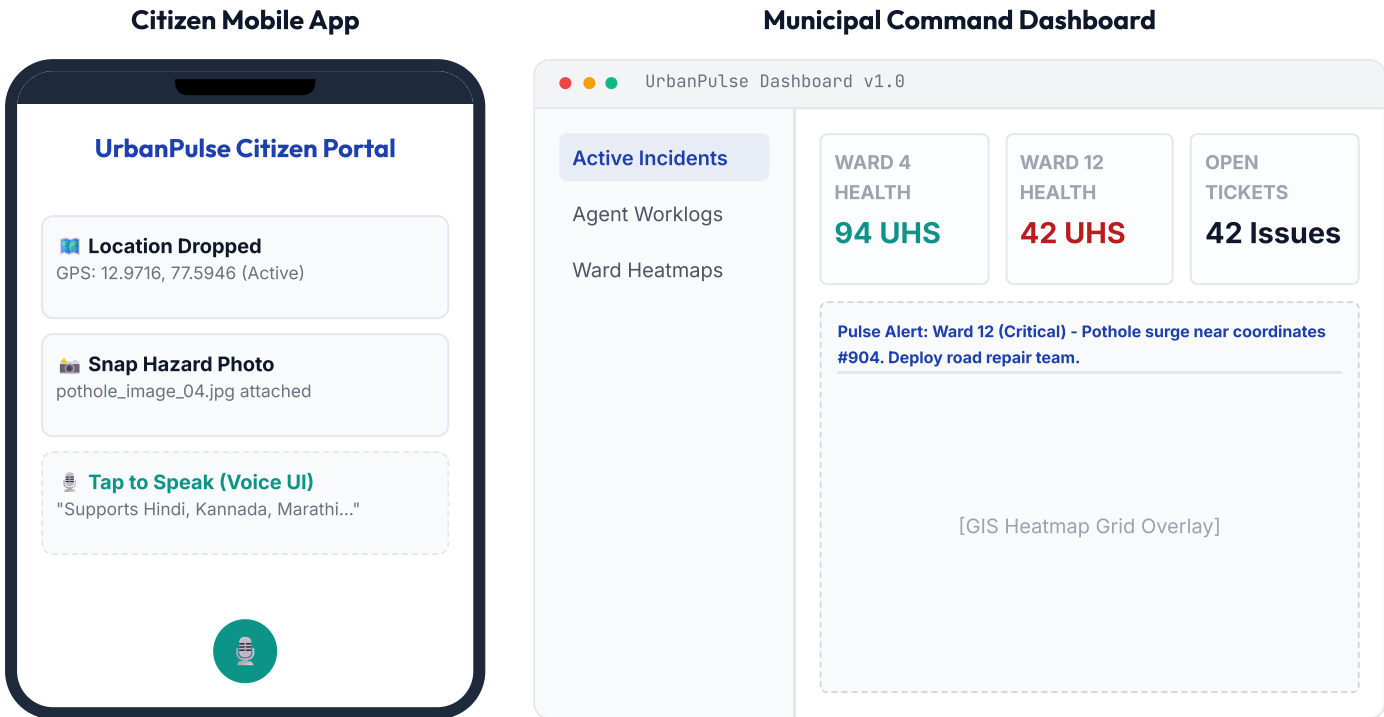
14. Evaluation Plan

To verify operational performance, the platform will be evaluated against key metrics:

- **Classification Accuracy:** Checked by verifying Vision Agent predictions against human-inspected Ground Truth categories.
- **Duplicate Detection Precision:** Measured via false-positive/false-negative deduplication matches within spatial buffer margins.
- **Average Routing Latency:** Tracked from initial report ingestion to final work order queue delivery.
- **SLA Compliance Rate:** Percentage of high-priority tickets resolved within target time limits.
- **Citizen Satisfaction Score:** Calculated from post-resolution follow-up ratings collected by the Citizen Experience Agent.
- **Dashboard Load Speeds:** Real-time PostGIS query response times in the Command Cockpit.

15. UI Mockup Wireframes

Below are the structural layouts for the Citizen Mobile App and the Municipal Cockpit Dashboard (showing the UHS and AI City Pulse widgets):



16. Conclusion & Scalability

UrbanPulse AI moves modern urban planning from a reactive ticket logger to an autonomous, context-aware city brain. By delegating individual cognitive workloads to collaborating, specialized AI agents, the platform automates triage, filters duplications, prioritizes community impact, and foresees structural failure. UrbanPulse AI is the ideal smart city solution for the Lenovo LEAP Hackathon, demonstrating a real-world, highly impactful application of multi-agent gen-AI systems.

Scalability Strategy

The platform is designed to scale horizontally across municipal divisions:

- **Multi-City Configuration:** Department routing maps are defined via standard YAML schemas, enabling quick config updates for deployment to new municipalities without modifying core code.
- **Cloud-Native Design:** All services are containerized, allowing agents and queue workers to scale dynamically on Kubernetes based on incoming report volumes.